

Equations with One Variable

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In the following question, I recommend that you solve using all possible methods for practice

1. Calculate the location of conduction with respect to the Fermi-Level in an n -type semiconductor with a doping of $N_D = 10^{20} \text{ cm}^{-3}$ using Maxwell-Boltzmann statistics. Repeat the calculations with the Fermi-Dirac statistics.
2. Calculate the location of conduction with respect to the Fermi-Level in an n -type semiconductor with a doping of $N_D = 10^{16} \text{ cm}^{-3}$ using Maxwell-Boltzmann statistics. Repeat the calculations with the Fermi-Dirac statistics.
3. Compare and comment on the results of Q1 and Q2
4. Calculate the energy at which the carrier concentration ($n(E)$) peaks i.e. $g_{3D}(E)f(E)$ is maximum

$$f(E) = \frac{1}{1 + \exp((E - E_F)/(k_B T))}$$
$$g_{3D} = \frac{1}{2\pi^2} \frac{2m^*}{\hbar^2} (E - E_C)^{0.5}$$

for the case of $N_D = 10^{18} \text{ cm}^{-3}$

5. Calculate the interatomic distance of NaCl given that the potential energy between two ions is

$$F(r) = -\frac{e^2}{4\pi\epsilon_0 r} + V_0 e^{r/r_0}$$

$V_0 = 1.09 \times 10^3 \text{ eV}$ and $r_0 = 0.33 \text{ Ang}$

6. In a piece of an n -type semiconductor the equation for carrier concentration (electron) accounting for incomplete ionization is given by

$$\frac{N_D N_C}{N_C + 2n e^{(E_C - E_D)/k_B T}} + \frac{N_C N_V}{n} e^{-E_G/k_B T} = n$$

Assume $E_C - E_D = 10 \text{ meV}$ $E_G = 1.1 \text{ eV}$. Calculate the carrier concentration at $T = 50 \text{ K}$, 300 K and 600 K . Analyze your results. From where do we get the above equation – From Charge neutrality condition.

7. A MOS Capacitor is a combination of two capacitors oxide and semiconductor in series. Manipulating the Poisson's equation for the MOS structure allows us to write the sheet charge density (C/cm^2) at the oxide-semiconductor interface as

$$\left| \frac{Q_s}{\epsilon_S} \right| = \left[\frac{\sqrt{2} k_B T}{q L_D} \right] \left[\left(e^{\frac{-q \phi_S}{k_B T}} + \frac{q \phi_S}{k_B T} - 1 \right) + \frac{n_0}{p_0} \left(e^{\frac{q \phi_S}{k_B T}} - \frac{q \phi_S}{k_B T} - 1 \right) \right]^{0.5}$$

where $L_D = \sqrt{\frac{\epsilon_S k_B T}{e^2 N_A}}$ and ϕ_S is the potential drop across the semiconductor. Calculate the potential drop across the semiconductor when $Q_S = 10^{-7} \text{ C/cm}^{-2}$, $10^{-8} \text{ C/cm}^{-2}$. Assume $N_A = 4 \times 10^{15} \text{ cm}^{-3}$, $T = 300 \text{ K}$